

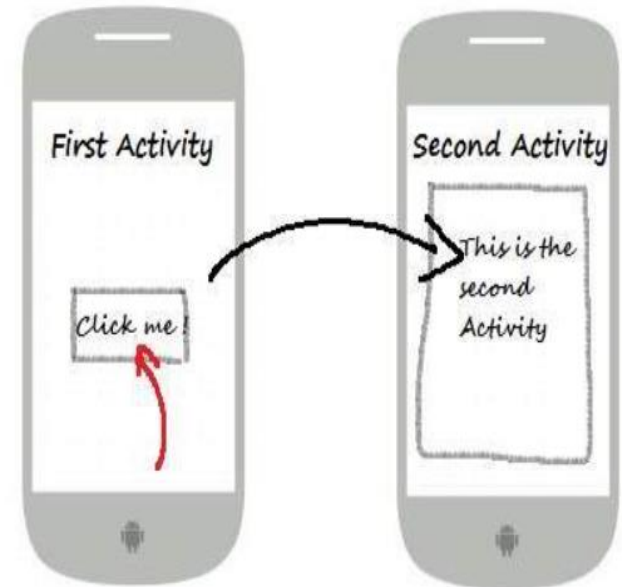
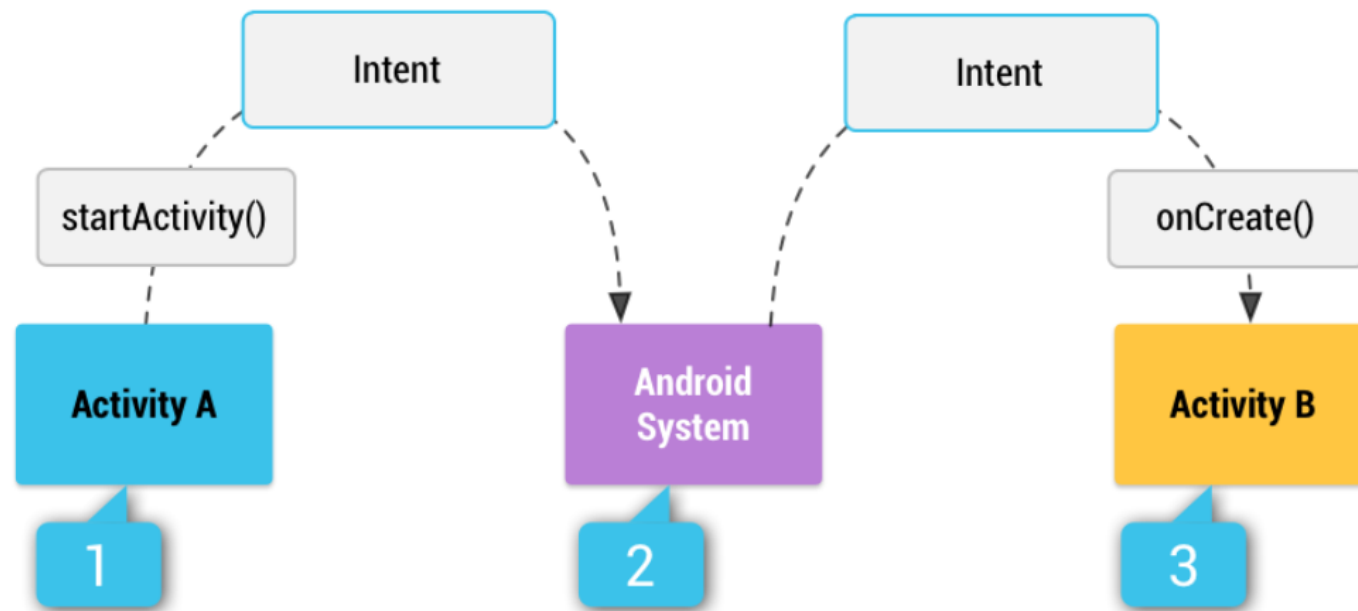
Introduction to Intents in Android App

What is Intent in Android?

- ▶ An Intent is a simple message object that is used to communicate between android components such as activities, content providers, broadcast receivers and services. Intents are also used to transfer data between activities.
- ▶ Intents are used generally for starting a new activity using `startActivity()`.

For example, if you want to send some message from Faisalabad to Karachi using the Post Office facility then you can do so by buying an Envelope and then pass the message in the Envelope and send the message to the desired location.

What is Intent in Android?



Use of Intent

- ▶ For Launching an Activity
- ▶ To start a New Service
- ▶ For Broadcasting Messages
- ▶ To Display a list of contacts in ListView

Types of Intent in Android

Intent is of two types:

- ▶ Implicit Intent
- ▶ Explicit Intent

Implicit Intent

- ▶ The implicit intent is the intent where instead of defining the exact components, you define the action that you want to perform for different activities.
- ▶ By using the Implicit Intents you can communicate between various applications present in the mobile device.
- ▶ Using an Implicit Intent is useful when your app cannot perform the action but other apps probably can and you'd like the user to pick which app to use.

Implicit Intent

For example, if you want to send images then the Android System will provide will options of Gmail, Instagram, Fb, WhatsApp, etc.

Syntax:

```
Intent i=new Intent();  
i.setAction(Intent.ACTION_SEND);
```

- There are some other standard actions that intents can use for launching activities.

Explicit Intent

- ▶ An explicit intent is an Intent where you explicitly define the component that needs to be called by the Android System.
- ▶ An explicit intent is one that you can use to launch a specific app component, such as a particular activity or service in your app.

For example, if you want to launch an Activity by clicking some button on the present Activity then you can specify the fully-qualified address of the desired Activity to launch that Activity.

- ▶ you can use Explicit Intents to have communication in your own application.

Explicit Intent



Explicit Intent

Syntax:

```
Intent I = new Intent(getApplicationContext(),NextActivity.class);  
I.putExtra("value1" , "This value for Next Activity");  
I.putExtra("value2" , "This value for Next Activity");
```

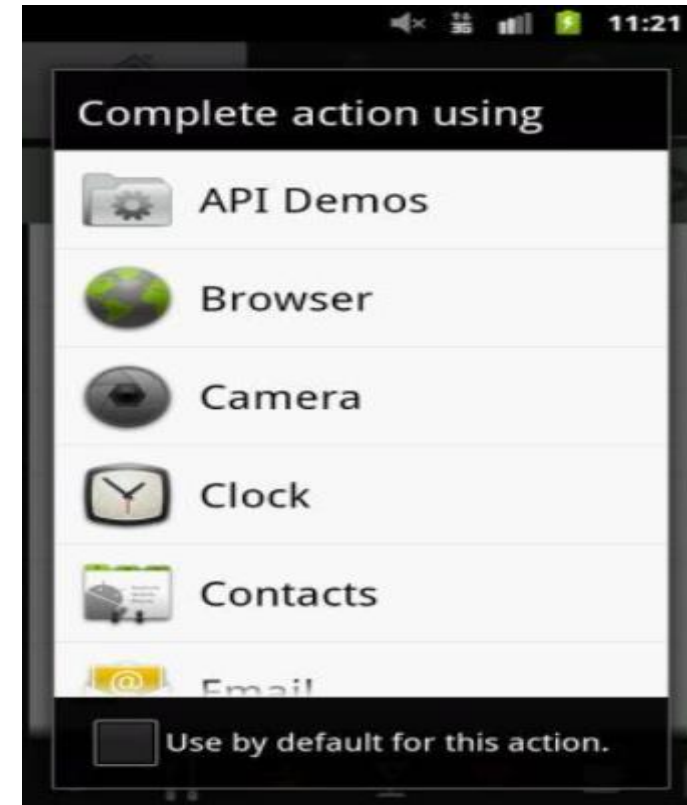
The Different Methods Used in Intent

- ▶ Action_Main
- ▶ Action_Pick
- ▶ Action_Chooser
- ▶ Action_Dial
- ▶ Action_Call
- ▶ Action_Send
- ▶ Action_SendTo
- ▶ Action_View

ACTION_MAIN

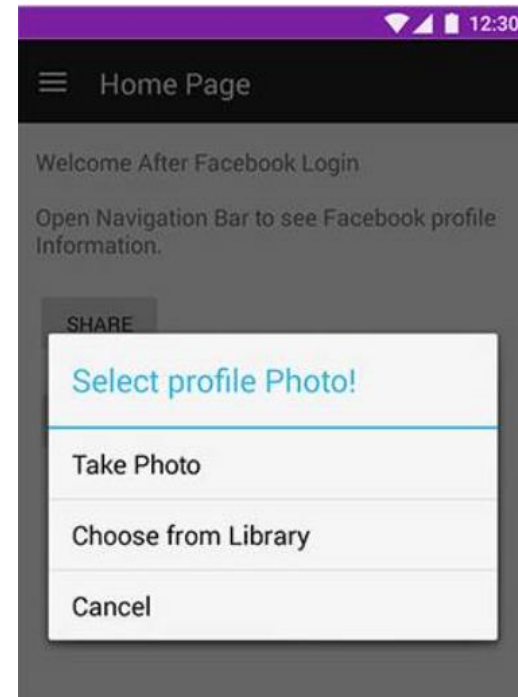
- Use: Adds an action to an intent Filter.

`<action android:name = "string">`



ACTION_PICK

- It is using for picking the image from CAMERA or GALLERY.



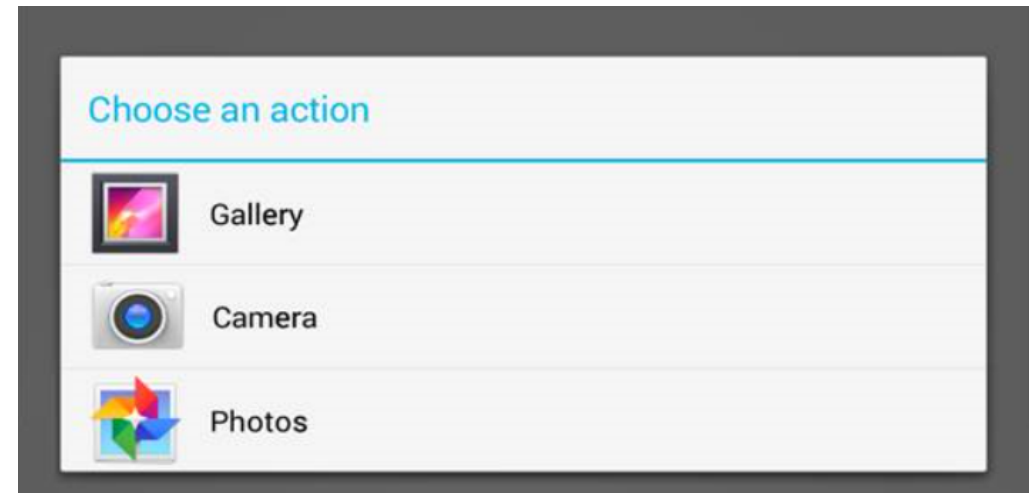
ACTION_PICK

► Syntax:

```
Intent photoPickerIntent = new Intent(Intent.ACTION_PICK);  
photoPickerIntent.setType("image/*");  
startActivityForResult(photoPickerIntent, SELECT_PHOTO);
```

ACTION_CHOOSER

- Use: It is used for choosing the image from the gallery.



ACTION_CHOOSER

- ▶ `Intent select = new Intent(Intent.ACTION_CAMERA);`
- ▶ `send.setData(uri);`
- ▶ `startActivity(Intent.createChooser(select, "Select an Image from Camera"));`

ACTION_DIAL

- Use – Display the phone dialer with the given number filled in.



ACTION_DIAL

```
String myPhoneNumber = "tel : 123456 ;  
Intent next = new Intent (Intent.ACTION_DIAL, Uri.parse(myPhoneNumberUri));  
startActivity(next);
```

ACTION_CALL

- Use: Placing and immediate phone call



ACTION_CALL

```
String data = "tel : 651234567";
```

```
Intent myActivity = new Intent(Intent.ACTION_CALL,Uri.parse(myNumber));
```

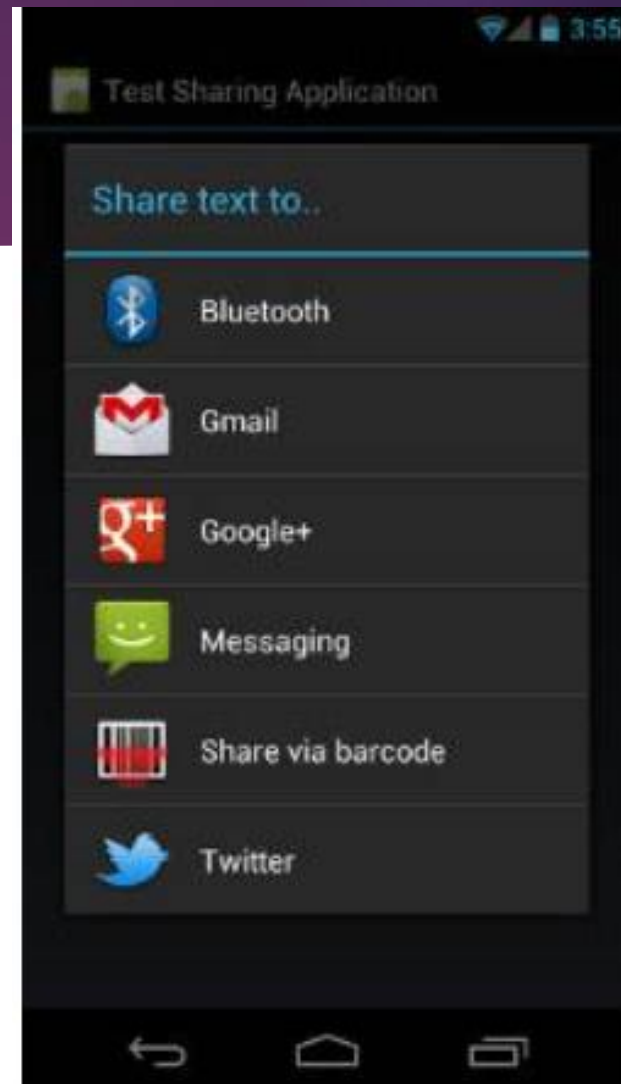
```
startActivity(myActivity);
```

Permission Needed:

```
<uses-permission android:name = "android.permission.CALL_PHONE" />
```

ACTION_SEND

- Use: Sending Text content from one activity to other.



ACTION_SEND

```
Intent sendIntent = new Intent();  
sendIntent.setAction(Intent.ACTION_SEND);  
sendIntent.putExtra(Intent.EXTRA_TEXT, "This is my text to Send");  
sendIntent.setType("text/plain");  
startActivity(sendIntent);
```

ACTION_SENDTO

- Use : Preparing an SMS. The text is supplied as an Extra element. The intent expects such as values to be called "sms_body"



ACTION_SENDTO

```
Intent intent = new Intent(Intent.ACTION_SENDTO, Uri.parse("smsto : 12345678 "));  
intent.putExtra("sms_body", "BS(CS)7th Evening is a Good Class? And  
Morning session is also awesome!");  
startActivity(intent);
```